

Lecture 3

Regularization and Evaluation

Machine Learning I

Alexander Quispe

ENEI – INEI · PEU-CD 2026

September 14, 2026

Outline

Ridge Regression

Penalties and Constraints

The Lasso

Choosing λ

Evaluating a Classifier

Summary

Two Loose Ends From Lectures 1–2

1. Least squares has variance $\sigma^2 \frac{p+1}{N}$, growing with every feature. When p approaches N — or when columns are nearly collinear — $(\mathbf{X}^\top \mathbf{X})^{-1}$ explodes and so does $\hat{\beta}$.
2. Logistic regression on separable data has no MLE: $\|\hat{\beta}\| \rightarrow \infty$.

Both are the same disease: nothing in the objective stops β from being large.

The cure

Add a penalty on the size of β to the empirical risk:

$$\hat{\beta}_\lambda = \arg \min_{\beta} \hat{R}(\beta) + \lambda \Omega(\beta), \quad \lambda \geq 0.$$

Today: $\Omega = \|\beta\|_2^2$ (ridge), $\Omega = \|\beta\|_1$ (lasso), what each does to the solution and why, how to pick λ , and — second half — how to score a classifier at all.

Ridge Regression

The Objective and Its Closed Form

$$J_\lambda(\boldsymbol{\beta}) = \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda \|\boldsymbol{\beta}\|^2 = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + \lambda \boldsymbol{\beta}^\top \boldsymbol{\beta}.$$

Gradient, then normal equations

Reuse Lecture 1 for the first term; the second is a quadratic form with $\mathbf{A} = \lambda \mathbf{I}$:

$$\nabla_{\boldsymbol{\beta}} J_\lambda = -2\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) + 2\lambda \boldsymbol{\beta} = 2[(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})\boldsymbol{\beta} - \mathbf{X}^\top \mathbf{y}].$$

Setting it to zero,

$$\hat{\boldsymbol{\beta}}_\lambda = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$$

- $\lambda = 0$ recovers least squares. $\lambda \rightarrow \infty$ drives $\hat{\boldsymbol{\beta}}_\lambda \rightarrow \mathbf{0}$.
- The Hessian is $2(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})$; next slide shows it is positive definite for every $\lambda > 0$, so the solution is unique — *regardless of the rank of $\mathbf{X}$* .

Invertibility

For $\lambda > 0$, $\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}$ is positive definite, hence invertible

For any $\mathbf{v} \neq \mathbf{0}$,

$$\mathbf{v}^\top (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}) \mathbf{v} = \|\mathbf{X}\mathbf{v}\|^2 + \lambda \|\mathbf{v}\|^2 \geq \lambda \|\mathbf{v}\|^2 > 0.$$

A symmetric matrix with all quadratic forms positive has all eigenvalues positive, so it is invertible. □

- Each eigenvalue of $\mathbf{X}^\top \mathbf{X}$ is shifted up by λ : $\lambda_j(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}) = \lambda_j(\mathbf{X}^\top \mathbf{X}) + \lambda$. Zero eigenvalues (collinearity, $p > N$) become λ .
- This is the origin of the name: the diagonal of $\mathbf{X}^\top \mathbf{X}$ gets a “ridge”.

Ridge always has a unique solution, including when $p > N$. That alone justifies it.

Ridge Is Biased

$$\mathbb{E}[\hat{\beta}_\lambda] \neq \beta$$

Under $\mathbf{y} = \mathbf{X}\beta + \varepsilon$, $\mathbb{E}[\varepsilon] = \mathbf{0}$,

$$\mathbb{E}[\hat{\beta}_\lambda] = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbb{E}[\mathbf{y}] = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{X} \beta = (\mathbf{I} + \lambda (\mathbf{X}^\top \mathbf{X})^{-1})^{-1} \beta.$$

Variance

$$\text{Var}(\hat{\beta}_\lambda) = \sigma^2 (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \preceq \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} = \text{Var}(\hat{\beta}_{\text{OLS}}).$$

Bias in, variance out. Whether the trade is worth it depends on λ — and there always exists a $\lambda > 0$ for which the total mean squared error of $\hat{\beta}_\lambda$ is below that of OLS (Hoerl & Kennard, 1970). The proof is in the homework; the mechanism is on the next two slides.

Ridge in the SVD Basis

Write the thin SVD $\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$: $\mathbf{U} \in \mathbb{R}^{N \times p}$ orthonormal columns, $\mathbf{D} = \text{diag}(d_1, \dots, d_p)$, $\mathbf{V} \in \mathbb{R}^{p \times p}$ orthogonal. Then $\mathbf{X}^\top \mathbf{X} = \mathbf{V}\mathbf{D}^2\mathbf{V}^\top$.

The fitted values, direction by direction

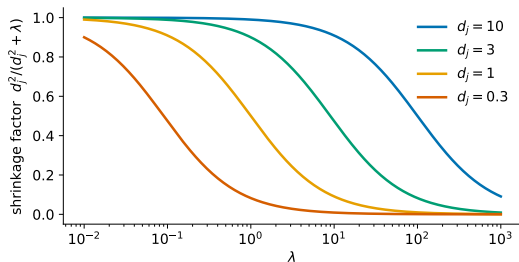
$$\begin{aligned}\hat{\mathbf{y}}_\lambda = \mathbf{X}\hat{\boldsymbol{\beta}}_\lambda &= \mathbf{U}\mathbf{D}\mathbf{V}^\top (\mathbf{V}\mathbf{D}^2\mathbf{V}^\top + \lambda\mathbf{V}\mathbf{V}^\top)^{-1} \mathbf{V}\mathbf{D}\mathbf{U}^\top \mathbf{y} & (\mathbf{I} = \mathbf{V}\mathbf{V}^\top) \\ &= \mathbf{U}\mathbf{D}\mathbf{V}^\top \mathbf{V}(\mathbf{D}^2 + \lambda\mathbf{I})^{-1} \mathbf{V}^\top \mathbf{V}\mathbf{D}\mathbf{U}^\top \mathbf{y} = \mathbf{U}\mathbf{D}(\mathbf{D}^2 + \lambda\mathbf{I})^{-1} \mathbf{D}\mathbf{U}^\top \mathbf{y} \\ &= \sum_{j=1}^p \mathbf{u}_j \frac{d_j^2}{d_j^2 + \lambda} \mathbf{u}_j^\top \mathbf{y}.\end{aligned}$$

Compare least squares: $\hat{\mathbf{y}} = \mathbf{U}\mathbf{U}^\top \mathbf{y} = \sum_j \mathbf{u}_j \mathbf{u}_j^\top \mathbf{y}$.

Ridge projects $\mathbf{y}$ onto each principal direction $\mathbf{u}_j$ of $\mathbf{X}$, then **shrinks** the coordinate by $\frac{d_j^2}{d_j^2 + \lambda} \in (0, 1)$.

Directions with small d_j — low variance in the inputs, high variance in $\hat{\boldsymbol{\beta}}$ — are shrunk the most. That is precisely where OLS variance lives.

Shrinkage Factors and Effective Degrees of Freedom



The trace of the ridge hat matrix $\mathbf{H}_\lambda = \mathbf{X}(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top$ generalizes Lecture 1's $\text{tr } \mathbf{H} = p$:

$$\text{df}(\lambda) = \text{tr } \mathbf{H}_\lambda = \sum_{j=1}^p \frac{d_j^2}{d_j^2 + \lambda}, \quad \text{df}(0) = p, \quad \text{df}(\infty) = 0.$$

A fractional number of parameters. It is the right x-axis for a bias–variance plot.

Two Practical Rules

Do not penalize the intercept. Shrinking β_0 toward 0 makes the fit depend on where the origin of y happens to be. Center $\mathbf{y}$ and the columns of $\mathbf{X}$ first; then $\hat{\beta}_0 = \bar{y}$ and the ridge problem is solved on the p slopes only.

Standardize the features. $\|\beta\|^2$ treats all coordinates alike; a feature measured in metres and the same feature in millimetres get coefficients differing by 10^3 and penalties differing by 10^6 . Divide each column by its standard deviation before fitting, and undo it afterwards.

Both rules apply verbatim to the lasso.

Ridge as a Bayesian Posterior Mode

Take the Gaussian model of Lecture 1 and add a prior on β :

$$\mathbf{y} \mid \beta \sim \mathcal{N}(\mathbf{X}\beta, \sigma^2 \mathbf{I}), \quad \beta \sim \mathcal{N}(\mathbf{0}, \tau^2 \mathbf{I}).$$

Maximize the log posterior

$$\begin{aligned} \log p(\beta \mid \mathbf{y}) &= \log p(\mathbf{y} \mid \beta) + \log p(\beta) + \text{const} \\ &= -\frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{X}\beta\|^2 - \frac{1}{2\tau^2} \|\beta\|^2 + \text{const} = -\frac{1}{2\sigma^2} \left[\|\mathbf{y} - \mathbf{X}\beta\|^2 + \frac{\sigma^2}{\tau^2} \|\beta\|^2 \right] + \text{const}. \end{aligned}$$

Maximizing over β is minimizing J_λ with $\lambda = \sigma^2/\tau^2$.

- A tight prior (small τ) is a large λ . The penalty encodes a belief that coefficients are small.
- The same computation with a Laplace prior $p(\beta_j) \propto e^{-|\beta_j|/b}$ gives the lasso, $\lambda = 2\sigma^2/b$.
- This is the cleanest answer to “where does the penalty come from”. It is also why the two are called ℓ_2 and ℓ_1 regularization: they are the log-densities of the two priors.

Penalties and Constraints

Two Formulations

Penalized

$$\min_{\beta} L(\beta) + \lambda \Omega(\beta)$$

One parameter $\lambda \geq 0$.

Constrained

$$\min_{\beta} L(\beta) \quad \text{s.t.} \quad \Omega(\beta) \leq t$$

One parameter $t \geq 0$.

Equivalence

For convex L and Ω : every solution of the penalized problem with parameter λ solves the constrained problem with $t = \Omega(\hat{\beta}_{\lambda})$; and every solution of the constrained problem with $t > 0$ solves the penalized problem for some $\lambda \geq 0$.

The map $\lambda \leftrightarrow t$ is monotone decreasing but depends on the data — one cannot convert without solving. The constrained form is the one to *draw*; the penalized form is the one to *compute*.

Equivalence: Both Directions

Penalized $\Rightarrow$ constrained (no convexity needed)

Let $\hat{\beta}_\lambda$ minimize $L + \lambda\Omega$ and set $t = \Omega(\hat{\beta}_\lambda)$. For any β with $\Omega(\beta) \leq t$,

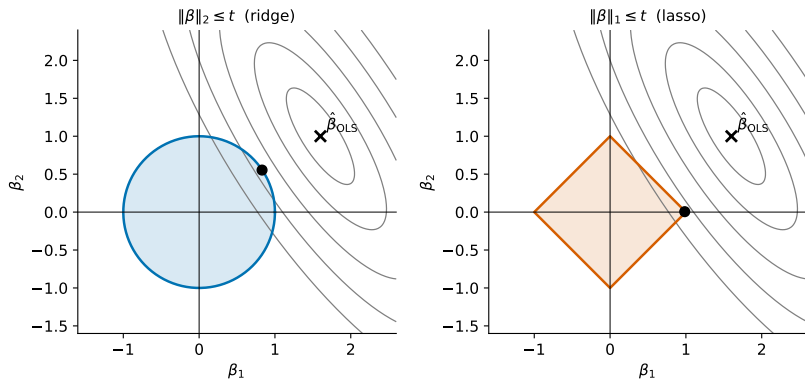
$$L(\beta) \geq L(\hat{\beta}_\lambda) + \lambda[\Omega(\hat{\beta}_\lambda) - \Omega(\beta)] = L(\hat{\beta}_\lambda) + \lambda[t - \Omega(\beta)] \geq L(\hat{\beta}_\lambda),$$

the first inequality being the optimality of $\hat{\beta}_\lambda$ for the penalized objective. So $\hat{\beta}_\lambda$ is feasible and no feasible point beats it. $\square$

Constrained $\Rightarrow$ penalized (convexity used here)

The constrained problem is convex with a strictly feasible point ($\beta = \mathbf{0}$, if $t > 0$), so Slater's condition holds and strong duality applies: there is a multiplier $\lambda^* \geq 0$ such that the constrained minimizer $\hat{\beta}_t$ minimizes the Lagrangian $L(\beta) + \lambda^*(\Omega(\beta) - t)$ over all β . The $-\lambda^*t$ is a constant, so $\hat{\beta}_t$ minimizes $L + \lambda^*\Omega$. $\square$

The Picture



Contours of $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$ (ellipses centred at $\hat{\boldsymbol{\beta}}_{OLS}$) meeting the constraint set $\Omega(\boldsymbol{\beta}) \leq t$. The ℓ_2 ball is round: the first touch is almost never on an axis. The ℓ_1 ball has corners on the axes: the first touch often is. The next section makes “often” precise.

The Lasso

The Objective

$$J_{\lambda}(\boldsymbol{\beta}) = \frac{1}{2} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda \|\boldsymbol{\beta}\|_1, \quad \|\boldsymbol{\beta}\|_1 = \sum_{j=1}^p |\beta_j|.$$

(The $\frac{1}{2}$ is a convention that removes a factor of 2 from every formula below.)

- Convex: a convex quadratic plus a norm. Every local minimum is global.
- Not differentiable wherever some $\beta_j = 0$. No closed form for general $\mathbf{X}$.
- Solutions are typically **sparse**: many $\hat{\beta}_j$ exactly 0. Ridge never does this.

Subgradient of the absolute value

$$\partial |\beta| = \begin{cases} \{+1\} & \beta > 0 \\ [-1, +1] & \beta = 0 \\ \{-1\} & \beta < 0 \end{cases} \quad \text{and } \boldsymbol{\beta}^* \text{ minimizes convex } J \iff \mathbf{0} \in \partial J(\boldsymbol{\beta}^*).$$

The Orthonormal Case: Soft Thresholding

Suppose $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$ (columns orthonormal). Write $\mathbf{b} = \mathbf{X}^\top \mathbf{y}$; note $\mathbf{b} = \hat{\beta}_{\text{OLS}}$.

The problem separates across coordinates

$$\frac{1}{2} \|\mathbf{y} - \mathbf{X}\beta\|^2 = \frac{1}{2} \|\mathbf{y}\|^2 - \mathbf{b}^\top \beta + \frac{1}{2} \beta^\top \underbrace{\mathbf{X}^\top \mathbf{X}}_{\mathbf{I}} \beta \Rightarrow J_\lambda(\beta) = \text{const} + \sum_{j=1}^p \left[\frac{1}{2} \beta_j^2 - b_j \beta_j + \lambda |\beta_j| \right].$$

Minimize one coordinate: $\min_{\beta} \frac{1}{2} \beta^2 - b\beta + \lambda |\beta|$

Stationarity: $0 \in \beta - b + \lambda \partial |\beta|$, i.e. $\beta = b - \lambda s$ for some $s \in \partial |\beta|$.

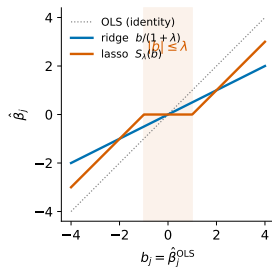
- If $\beta > 0$: $s = 1$, $\beta = b - \lambda$; consistent iff $b > \lambda$.
- If $\beta < 0$: $s = -1$, $\beta = b + \lambda$; consistent iff $b < -\lambda$.
- If $\beta = 0$: need $0 = b - \lambda s$ with $s \in [-1, 1]$, i.e. $|b| \leq \lambda$.

The three cases partition $\mathbb{R}$. Hence

$$\hat{\beta}_j = S_\lambda(b_j) \equiv \text{sign}(b_j) (|b_j| - \lambda)_+$$

Soft Thresholding vs. Ridge Shrinkage

In the same orthonormal setting, ridge gives $\hat{\beta}_j = (\mathbf{I} + \lambda \mathbf{I})^{-1} b_j = \frac{b_j}{1 + \lambda}$.



Ridge: multiply every coefficient by the same factor < 1 . Zero stays zero; nothing else becomes zero.

Lasso: subtract λ from every $|b_j|$, floor at 0. Everything with $|b_j| \leq \lambda$ is set to exactly zero.

Numerical Example

Orthonormal design, $\mathbf{b} = \hat{\beta}_{\text{OLS}} = (3, 0.5, -2)^\top$, $\lambda = 1$.

	$b_1 = 3$	$b_2 = 0.5$	$b_3 = -2$
Ridge $b_j/(1 + \lambda)$	1.5	0.25	-1
Lasso $\text{sign}(b_j)(b_j - \lambda)_+$	2	0	-1

Ridge halves everything. Lasso removes the small coefficient and shifts the large ones by a constant — so the large ones are *less* shrunk under lasso than under ridge here (2 vs 1.5).

Lasso bias is roughly constant in $|b_j|$; ridge bias is proportional to $|b_j|$. Lasso hurts large true coefficients less, and kills small ones outright.

General $\mathbf{X}$: When Is $\hat{\beta}_j$ Exactly Zero?

KKT conditions for the lasso

Stationarity, coordinate by coordinate, with $\mathbf{x}^{(j)}$ the j -th column of $\mathbf{X}$ and $\mathbf{r} = \mathbf{y} - \mathbf{X}\hat{\beta}$:

$$0 \in -\mathbf{x}^{(j)\top} \mathbf{r} + \lambda \partial |\hat{\beta}_j| \quad \Longleftrightarrow \quad \mathbf{x}^{(j)\top} \mathbf{r} = \lambda s_j, \quad s_j \in \partial |\hat{\beta}_j|.$$

Reading it:

- If $\hat{\beta}_j \neq 0$: $\mathbf{x}^{(j)\top} \mathbf{r} = \lambda \text{sign}(\hat{\beta}_j)$. Every *active* feature has correlation with the residual of magnitude exactly λ .
- If $\hat{\beta}_j = 0$: $|\mathbf{x}^{(j)\top} \mathbf{r}| \leq \lambda$. An *inactive* feature is one whose correlation with the residual does not reach λ .

A coefficient is zero iff its feature explains less than λ worth of the residual. There is a whole interval of λ values for which this holds, so zeros are not a coincidence: they have positive probability.

Corollary: $\hat{\beta} = \mathbf{0}$ iff $|\mathbf{x}^{(j)\top} \mathbf{y}| \leq \lambda$ for all j , i.e. iff $\lambda \geq \lambda_{\max} = \|\mathbf{X}^\top \mathbf{y}\|_\infty$. The path starts there.

Solving It: Coordinate Descent

Fix all coordinates but β_j . Let $\mathbf{r}_{(-j)} = \mathbf{y} - \sum_{k \neq j} \mathbf{x}^{(k)} \beta_k$ be the partial residual.

One coordinate is a one-dimensional lasso

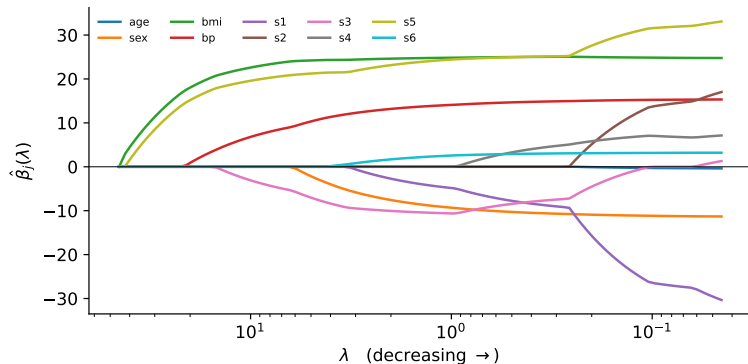
$$\frac{1}{2} \left\| \mathbf{r}_{(-j)} - \mathbf{x}^{(j)} \beta_j \right\|^2 + \lambda |\beta_j| = \frac{\left\| \mathbf{x}^{(j)} \right\|^2}{2} \beta_j^2 - (\mathbf{x}^{(j)\top} \mathbf{r}_{(-j)}) \beta_j + \lambda |\beta_j| + \text{const.}$$

Dividing by $\left\| \mathbf{x}^{(j)} \right\|^2$ puts it in the form of the orthonormal case, with $b = \mathbf{x}^{(j)\top} \mathbf{r}_{(-j)} / \left\| \mathbf{x}^{(j)} \right\|^2$ and threshold $\lambda / \left\| \mathbf{x}^{(j)} \right\|^2$:

$$\beta_j \leftarrow \frac{S_\lambda(\mathbf{x}^{(j)\top} \mathbf{r}_{(-j)})}{\left\| \mathbf{x}^{(j)} \right\|^2}$$

- Cycle $j = 1, \dots, p$ until nothing changes. Each update is $O(N)$; a full sweep is $O(Np)$.
- Compute the whole path by starting at $\lambda_{\max}$ and decreasing λ on a grid, warm-starting each solve from the previous one. This is what `glmnet` and `scikit-learn` do.

The Lasso Path



Coefficients of the diabetes data as λ decreases from $\lambda_{\max}$ (left) to 0 (right). Features enter one at a time, at the λ where their correlation with the residual reaches λ . The path is piecewise linear in λ — a fact that makes the whole thing computable exactly.

Beyond Ridge and Lasso

- **Elastic net:** $\lambda_1 \|\beta\|_1 + \lambda_2 \|\beta\|_2^2$. Keeps the sparsity of ℓ_1 , adds the stability of ℓ_2 when features are correlated (lasso tends to pick one of a correlated group at random).
- **Regularized logistic regression:** replace $\frac{1}{2} \|\mathbf{y} - \mathbf{X}\beta\|^2$ by the log loss. The objective is strictly convex for any $\lambda > 0$, so the separable-data problem of Lecture 2 disappears. Coordinate descent still works, via IRLS on each coordinate.
- **Early stopping** is regularization too: for gradient descent on least squares from $\beta = \mathbf{0}$, the iterate after t steps shrinks direction j by $1 - (1 - \eta d_j^2)^t$ — the same shape as ridge's $d_j^2 / (d_j^2 + \lambda)$, with t playing the role of $1/\lambda$.

Every method in this course has a knob that trades bias for variance. Ridge and lasso are the two whose knob has a closed-form meaning.

Choosing λ

Training Error Is Optimistic — By Exactly How Much

For any *linear smoother* $\hat{\mathbf{y}} = \mathbf{H}\mathbf{y}$ (least squares, ridge, and others) under $\mathbf{y} = \mathbf{f} + \varepsilon$, $\text{Var}(\varepsilon) = \sigma^2 \mathbf{I}$:

Expected training error vs. expected test error at the same inputs

Let $\mathbf{y}'$ be a fresh copy of the responses at the same $\mathbf{X}$. Then

$$\begin{aligned}\mathbb{E} \|\mathbf{y}' - \hat{\mathbf{y}}\|^2 - \mathbb{E} \|\mathbf{y} - \hat{\mathbf{y}}\|^2 &= 2 \sum_{i=1}^N \text{Cov}(y_i, \hat{y}_i) \\ &= 2 \sum_{i=1}^N \text{Cov}\left(y_i, \sum_k H_{ik} y_k\right) = 2\sigma^2 \sum_i H_{ii} = 2\sigma^2 \text{tr } \mathbf{H}.\end{aligned}$$

Second line: expand both squares; $\mathbf{y}' \perp \hat{\mathbf{y}}$ kills one cross term, and $\mathbb{E} \|\mathbf{y}'\|^2 = \mathbb{E} \|\mathbf{y}\|^2$.

$$\mathbb{E}[\text{test error}] = \mathbb{E}[\text{training error}] + \frac{2\sigma^2}{N} \text{df} \quad (\text{per observation, df} = \text{tr } \mathbf{H})$$

The optimism grows with the degrees of freedom. This is Mallows' C_p , and the reason $\text{df}(\lambda)$ was worth defining: it is the currency in which overfitting is paid.

Cross-Validation

When σ^2 and df are unknown — most of the time — estimate the test error directly.

Input: data $\mathcal{D}$, grid $\{\lambda_1, \dots, \lambda_m\}$, number of folds K

Split $\mathcal{D}$ at random into K folds $\mathcal{D}_1, \dots, \mathcal{D}_K$ of equal size;

for λ in the grid **do**

for $k = 1, \dots, K$ **do**

 fit $\hat{\beta}_\lambda^{(-k)}$ on $\mathcal{D} \setminus \mathcal{D}_k$; $e_k(\lambda) \leftarrow$ error of $\hat{\beta}_\lambda^{(-k)}$ on $\mathcal{D}_k$;

end

$CV(\lambda) \leftarrow \frac{1}{K} \sum_k e_k(\lambda)$, $se(\lambda) \leftarrow \text{s.d.}(e_1, \dots, e_K) / \sqrt{K}$;

end

Output: $\hat{\lambda} = \arg \min_\lambda CV(\lambda)$, or the largest λ with $CV(\lambda) \leq \min CV + se$

- $K = 5$ or 10 . Larger K : less bias (bigger training sets), more variance and cost.
- The **one-standard-error rule** picks the simplest model statistically indistinguishable from the best.
- Standardize *inside* each fold. Standardizing once on all of $\mathcal{D}$ leaks the held-out fold's scale into training — a small leak, but a leak.

Evaluating a Classifier

Accuracy Is Not Enough

A classifier outputs $\hat{y} \in \{0, 1\}$; the truth is $y \in \{0, 1\}$. Four outcomes:

	$\hat{y} = 1$	$\hat{y} = 0$
$y = 1$	true positive (TP)	false negative (FN)
$y = 0$	false positive (FP)	true negative (TN)

$$\text{Accuracy} = \frac{TP + TN}{N}, \quad \text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}.$$

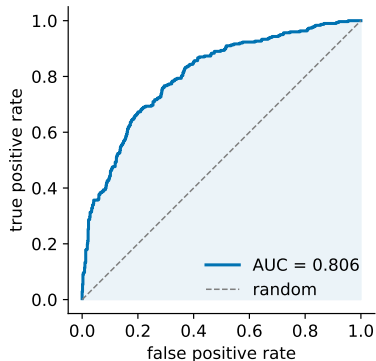
With 1% positives, predicting “0” for everyone has 99% accuracy, 0 recall, and undefined precision. Accuracy rewards agreeing with the majority. Precision and recall each measure one kind of mistake.

The harmonic mean $F_1 = 2 \frac{P \cdot R}{P + R}$ is the standard single summary: harmonic, not arithmetic, so that it is small whenever *either* is small.

Thresholds and the ROC Curve

Most classifiers produce a score $s(\mathbf{x})$ (a probability, a margin) and predict $\hat{y} = \mathbf{1}\{s(\mathbf{x}) \geq \tau\}$. Every τ gives a different confusion matrix. Sweep it:

$$\text{TPR}(\tau) = \frac{\text{TP}}{\text{TP} + \text{FN}} = \text{recall}, \quad \text{FPR}(\tau) = \frac{\text{FP}}{\text{FP} + \text{TN}}.$$



- $\tau \rightarrow \infty$: nothing positive, $(0, 0)$. $\tau \rightarrow -\infty$: everything positive, $(1, 1)$.
- A random score gives the diagonal. Perfect separation gives the corner $(0, 1)$.
- **AUC** = area under the curve. Threshold-free; invariant to monotone transforms of s .
- Precision–recall curves and **average precision** do the same with (recall, precision), and are preferable when positives are rare: FPR barely moves when TN is huge.

What AUC Actually Measures

AUC = $\mathbb{P}(s(\mathbf{x}^+) > s(\mathbf{x}^-))$ for a random positive $\mathbf{x}^+$ and a random negative $\mathbf{x}^-$

Proof for the empirical curve, N_+ positives and N_- negatives, distinct scores

Sort all scores. Moving τ down past a *negative* moves the curve right by $1/N_-$; past a *positive*, up by $1/N_+$. The curve is a staircase, and its area is the sum, over each negative, of the height of the curve when that negative is crossed:

$$\text{AUC} = \sum_{\text{neg } j} \frac{1}{N_-} \cdot \frac{\#\{\text{pos } i : s_i > s_j\}}{N_+} = \frac{1}{N_+ N_-} \sum_{i \in +} \sum_{j \in -} \mathbf{1}\{s_i > s_j\}. \quad \square$$

- This is the Mann–Whitney U statistic. AUC is a rank statistic, which is why it ignores calibration.
- An AUC of 0.5 means the scores carry no ranking information. Below 0.5 means flip the sign.

Costs Are Asymmetric: Choosing the Threshold

Suppose a false positive costs c_{FP} and a false negative costs c_{FN} , and the classifier outputs a calibrated $p(\mathbf{x}) = \mathbb{P}(y = 1 \mid \mathbf{x})$.

Minimize expected cost at $\mathbf{x}$

Predict 1 : $\mathbb{E}[\text{cost}] = (1 - p) c_{FP}$ (wrong iff $y = 0$)

Predict 0 : $\mathbb{E}[\text{cost}] = p c_{FN}$ (wrong iff $y = 1$)

Predict 1 iff $(1 - p) c_{FP} \leq p c_{FN}$, i.e.

$$\hat{y} = 1 \iff p(\mathbf{x}) \geq \tau^* = \frac{c_{FP}}{c_{FP} + c_{FN}}$$

- Equal costs: $\tau^* = \frac{1}{2}$ — the Bayes classifier.
- A false negative ten times worse than a false positive: $\tau^* = 1/11 \approx 0.09$. Predict positive on weak evidence, because missing one is expensive.
- The threshold is a property of the *decision*, not of the model. Train for probabilities (log loss), then choose τ from the costs. Never bake the costs into the training labels.

Multiclass, Briefly

For $K > 2$ classes the confusion matrix is $K \times K$, and each class k has its own precision and recall (one-vs-rest).

- **Macro** average: mean of the per-class scores. Every class counts equally, however rare.
- **Micro** average: pool all TP, FP, FN, then compute once. Equals accuracy for single-label problems.
- Report the full matrix when K is small. Most errors have structure — two classes confused with each other — that no average shows.

The metric is part of the problem statement, not of the method. Decide it before fitting anything.

Summary

What We Proved Today

1. Ridge: $\hat{\beta}_\lambda = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$, unique for all $\lambda > 0$; in the SVD basis it shrinks direction j by $d_j^2 / (d_j^2 + \lambda)$; $\text{df}(\lambda) = \sum_j d_j^2 / (d_j^2 + \lambda)$; it is the posterior mode under a Gaussian prior with $\lambda = \sigma^2 / \tau^2$.
2. Penalized and constrained formulations are equivalent (one direction elementary, the other via Slater).
3. Lasso with orthonormal $\mathbf{X}$: $\hat{\beta}_j = \text{sign}(b_j)(|b_j| - \lambda)_+$. In general, $\hat{\beta}_j = 0$ iff $|\mathbf{x}^{(j)\top} \mathbf{r}| \leq \lambda$; coordinate descent applies soft thresholding one coordinate at a time.
4. Optimism of training error $= 2\sigma^2 \text{df} / N$.
5. $\text{AUC} = \mathbb{P}(s^+ > s^-)$; the cost-optimal threshold is $c_{\text{FP}} / (c_{\text{FP}} + c_{\text{FN}})$.

Next

Lab 2 (Wednesday, Rodrigo) — ridge and lasso paths on real data, cross-validated λ with the one-SE rule, and what regularization does to logistic regression on separable data.

Lecture 4 (Friday) — a different hypothesis class altogether: decision trees. Impurity measures and why concavity matters, information gain, greedy training, and bagging — the first method that reduces variance without touching bias.

Reading — *ESL* §3.4 (ridge, lasso), §3.8.6 (coordinate descent), §7.4–7.5 (optimism, C_p), §7.10 (cross-validation). Tibshirani (1996) for the lasso in the original.